Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	02 February 2026
Team ID	LTVIP2026TMIDS34601
Project Name	TransLingua
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Text Translation	Translate text from source language to target language using Google Gemini AI Support maximum 3000 characters per translation request
FR-2	Auto Language Detection	Automatically detect the source language of input text Display detected language to the user before translation
FR-3	Multi-Language Support	Support 25+ languages including English, Spanish, French, German, Italian, etc.
FR-4	User Interface	Display futuristic cyberpunk-themed web interface Show real-time translation progress with progress bar
FR-5	Translation Controls	Provide Translate button to initiate translation Provide Swap button to switch source and target languages
FR-6	Sidebar Information	Display application features list Display use cases for the translator

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The application must provide an intuitive, user-friendly interface with clear labels, responsive layout, and visual progress feedback so that users with no technical expertise can translate text within seconds.
NFR-2	**Security**	API keys must be stored securely in environment variables (.env file) and never exposed in source code or client-side. The .gitignore must exclude .env to prevent accidental secret leaks. No user data shall be collected or stored.
NFR-3	**Reliability**	The application must gracefully handle API errors, network failures, and invalid inputs by showing meaningful error messages without crashing. The auto-detection feature must fallback to English if detection fails.
NFR-4	**Performance**	Translation requests must complete within 3 seconds for standard text (under 1000 characters). The UI must remain responsive during API calls with proper loading indicators.
NFR-5	**Availability**	The application shall be available 99.9% of the time when hosted, dependent on Google Gemini API uptime. Streamlit's built-in server handles concurrent user sessions.
NFR-6	**Scalability**	The architecture must support future additions of new languages, voice translation, document upload, and batch processing without major refactoring of the core translation module.